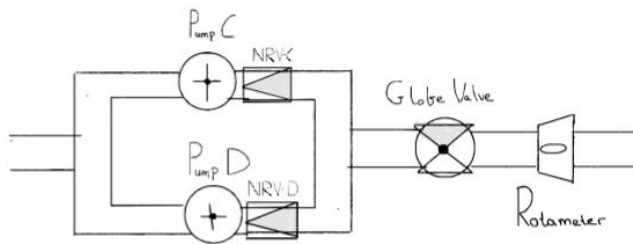


OBJECTIVE 1: PROPOSED SOLUTION TO CLIENT

1.1 System Schematic

Figure 1: Proposed Design Solution, Dual Centrifugal Pump Station (Pumps D & C in Parallel)



Key Components:

- **Pump C (D = 200 mm):** Centrifugal pump operating at fixed speed $N = 2900$ rpm.
- **Pump D (D = 250 mm):** Centrifugal pump operating at fixed speed $N = 2900$ rpm.
- **Non-Return (Check) Valve:** Installed at each pump discharge to prevent backflow when the system is shut down.
- **Manual Globe Valve:** Located on the combined discharge line for throttling of the system.
- **Flow Indicator (Rotameter):** Provides visual flow feedback without electronic systems.

1.2 Operating Instructions

To achieve any flow rate between 6 and 780 ltr/min, the client should follow these manual procedures:

- **Low-Flow Range (6–348 ltr/min):** Activate Pump C only. The NRV on the stationary Pump D will automatically prevent backflow.
- **High-Flow Range (349–780 ltr/min):** Activate both Pump C and Pump D in parallel. The combined head will overcome the 5m system resistance to meet demand.
- **Flow Setting:** Use the manual globe valve to throttle the discharge. This shifts the operating point along the performance curve until the target value is displayed on the rotameter.
- **Shutdown:** Close the globe valve before powering down motors to prevent system surge.

1.3 Design Justification

To meet the requirement of 6–780 ltr/min at a 5 m head, a parallel configuration of Pumps C and D is the necessary solution. This setup provides a combined flow of approximately 1064 ltr/min when operating at the 5 m system head, ensuring the peak capacity is met with a safety margin.

Pumps A and B are unsuitable because they cannot overcome the 5 m system head. Individually, Pump C and Pump D each fall short of the peak flow requirement of 780 ltr/min (providing only 348.6 ltr/min and 715.4 ltr/min respectively at 5m head), necessitating the parallel configuration.

The use of manual globe valve and mechanical rotameter ensures compliance with the no-electronic equipment constraint while also minimizing cost.

OBJECTIVE 2: PUMP PERFORMANCE CURVES & ANALYSIS

2.1 Graphs

Figure 2: Measured Pump a Lab Data at Variable Rotational Speeds (3300, 2700, 2100 RPM)

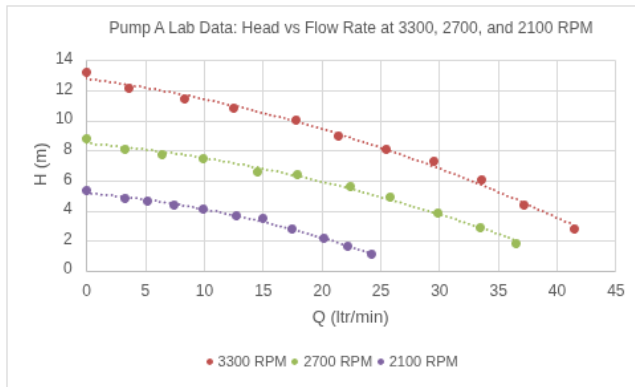


Figure 3: Predicted Performance Curves for Pumps A, B, C and D Scaled to N=2900

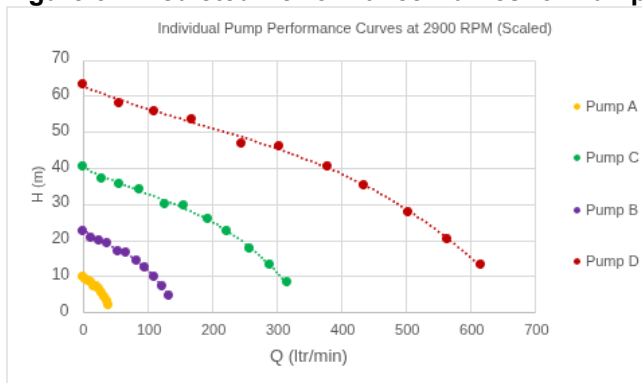
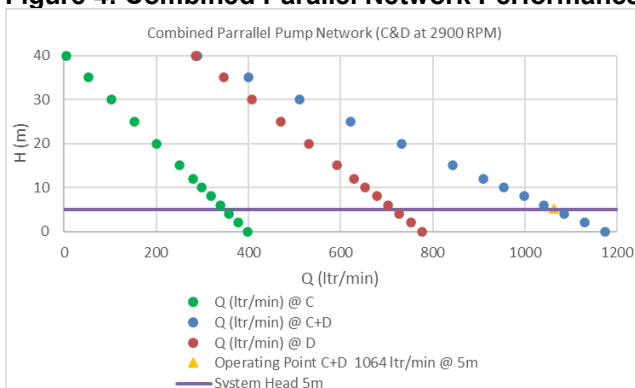


Figure 4: Combined Parallel Network Performance Curve for C and D at 2900 RPM



2.2 Performance Summary

Pump	D (mm)	Max Q (m³/s)	H @ Max Q (m)	Q @ 5 m (ltr/min)	Meets 5m Head?
A	100	0.000655185	2.117451111	-	No
B	150	0.00221125	4.764265	-	No
C	200	0.005241481	8.469804444	348.6	Yes
D	250	0.010237269	13.23406944	715.4	Yes
Combined	-	-	-	1064	Yes – selected